

The DR-2 programme – a chronicle from the carrier-sharing obstruction

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

This note chronicles the DR-2 programme end-to-end. DR-2 is the unrestricted sphere additive-energy bound $E_+(Q) \lesssim_\epsilon N^{2+\epsilon}$, the route to removing the angular-separation hypothesis H-ADM-COH from the B1 selection proof. The chronicle records what was tried, what failed, what closed, and at what tier, so the programme’s logic is auditable from one place. It introduces no new result; every claim is cited to its result-ledger entry.

2. The obstruction: DR2-SHARE

The elementary (extraction) route (dr2-extraction-lemmas v1.0) reduces the additive energy of a competitor pattern to a sum over “carrier” circles: the homogeneous part is capped per circle (universal $K = 14$), and the dichotomy adjoins rich circles. The sticking point, DR2-SHARE, is that adjoined circles SHARE points, so $\sum_s I_{C_s}^2 \leq I^2$ fails and the cap degrades with the sharing multiplicity $\chi(P) = \max_x \#\{\text{rich carriers through } x\}$. The elementary route closes iff $\chi = O(\text{polylog})$ class-wide – an incidence statement adjacent to a hard open conjecture.

3. Three routes

(I) Decoupling. Bourgain–Demeter ℓ^2 -decoupling at $p = 4$, $d = 3$ gives $E_+ \lesssim_\epsilon N^{2+\epsilon}$ for δ -separated Q (R-022, separated case T6 conditional). The non-separated case needs an induction-on-scales: the affine-invariance of additive energy (R-023, T7) plus a decoupling-iteration inequality (R-024). Two corrections were forced by audit (the false torus identity $\|f_\theta\|_4^4 = E_+(Q_\theta)$, repaired via the Besicovitch mean $M(|f_Q|^4) = E_+(Q)$; the “exact paraboloid” overclaim, softened to C^2 -perturbation stability). The honest residual is two cited ingredients (local-to-global averaging; multi-scale ϵ -bookkeeping – the hard core of decoupling), so the arbitrary- Q decoupling route rests at T4+ STRONG EVIDENCE.

(II) Elementary / PSM. The external research reduces DR-2 to the Pencil Rigidity / Pair-Sum Surface Multiplicity conjecture (T2), via the cross-energy lemma (R-021, T7 – which also corrected the external constant $2|A||B|$ to $2\min(\cdot)$). This is a sound conditional reduction, not a closure.

(III) Lattice (Route A) – the one that closed. The competitor patterns in TECT are not adversarial: they are BCC/FCC reciprocal-lattice momentum-shell points. The sum-level-circle lemma $E_+(Q) \leq (1 + T'(Q))N^2$ (R-025, T7 unconditional) reduces DR-2 to the sum-circle richness T' . For lattice shells, the substitution $y = 2x - t$ (the sum-circle centre is $t/2$) sends $\mathbb{Z}^3 \cap C_t$ injectively into $\{y \in \Lambda_t : |y|^2 = 4R - |t|^2\}$, a homogeneous rank-2 representation count $\leq 6d(4R -$

$|t|^2) \ll_\epsilon R^\epsilon$ by the Dirichlet class-number formula (uniform). Hence $E_+(Q) \leq (1 + C_\epsilon R^\epsilon)N^2$ for any crystallographic-shell subset (R-026, T7 UNCONDITIONAL modulo textbook number theory; decoupling-free).

4. The results ledger (chronological)

ID	Result	Tier
R-021	Sphere-circle cross-energy bound (corrected constant)	T7
R-022	DR-2 via ℓ^2 -decoupling (separated / unrestricted)	T6 cond / T4+
R-023	Affine-invariance of additive energy	T7
R-024	Decoupling-iteration inequality (Besicovitch-repaired)	T4+
R-025	Sum-level-circle bound $E_+ \leq (1 + T')N^2$	T7
R-026	DR-2 for the lattice class ([DIV-CIRC] proved)	T7
R-027	Weighted $G1'''$ -AE bridge $\sum w_t ^2 \leq (1 + T')I^2$	T7
R-028	H-ADM-COH discharge decision (finite margin)	enacted

5. The resolution and the discharge

The weighted bridge (R-027, T7): for any complex amplitudes c , $\sum_t |w_t|^2 \leq (1 + T'(Q))I^2$, $I = \sum_a |c_a|^2$. Identifying the STEP-5B corner $G1'''$ -AE with this weighted additive energy, R-026 gives, for the lattice class, $\sum_t |w_t|^2 \leq (1 + C_\epsilon R^\epsilon)I^2$ – the $G1'''$ -AE bound with NO angular separation. The acceptance question “is a subpolynomial $K \sim R^\epsilon$ tolerable?” was settled (R-028) as the finite inequality $K_{\text{adm}} = 1 + T'(Q) \leq K_{\text{allowed}}(n) = 8 + 4\sqrt{14}\sqrt{n}$, verified with margin $10.6 \times -15.8 \times$ (growing in n). Since H-ADM-COH only secured $G1'''$ -AE by angular separation, the operator (2026-06-08) DISCHARGED it for the crystallographic lattice class, reducing the B1 active hypothesis set $\{H\text{-LAYER}, H\text{-ADM-COH}, SC\text{-SCOPE}\} \rightarrow \{H\text{-LAYER}, SC\text{-SCOPE}\}$ with B1 tier UNCHANGED T6.

6. What remains open

The arbitrary real-point DR-2 ($Q \subset S^2$ not a lattice subset) is NOT closed; it rests at T4+ via the decoupling route (residuals R1/R2) and is retained only as a legacy fallback. It is not needed for the TECT mainline, which uses only the crystallographic lattice class. The PSM conjecture (T2) remains an alternative open route.

7. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “A chronicle could overstate by blurring the lattice closure with the open arbitrary- Q problem.” Guarded: §3(III) and §5 scope the closure to the lattice class explicitly, and §6 keeps arbitrary- Q OPEN.
- (β) “It might claim B1 was promoted.” It does not: B1 tier is UNCHANGED T6; only the hypothesis set shrank (a strengthening of the conditional, not a tier lift).
- (γ) “It might present the decoupling route as closed.” No: R-022/R-024 are recorded at T4+ with the two cited residuals named.

Quantitative sanity check (mandatory). All numbers are cited to their verified result entries: $T'/N \rightarrow 0$ (R-025/R-026, slope 0.18); $6d$ bound and substitution exact (R-026, `dr2_divcirc_proof 4/4`); weighted bound worst ratio 0.277 (R-027); margin $10.6\text{--}15.8\times$ (R-028). No new number is asserted.

8. Result footer

Result ID:	B5 / DR-2 programme consolidation (chronicle)
Precise statement:	A one-place chronicle of the DR-2 programme: obstruction (DR2-SHARE, $\chi(P)$); three routes (decoupling T4+, elementary/PSM T2, lattice T7); results R-021..R-028; resolution (lattice $E_+ \leq (1+C R^\epsilon)N^2$ + weighted bridge + finite margin) and the operator-enacted H-ADM-COH discharge for the crystallographic class.
Scope:	Narrative consolidation of B5 DR-2; no new claim.
Dependencies:	R-021..R-028 (each cited at its own tier).
Evidence grade:	CHRONICLE (no independent result). Final state: G1'''-AE_lattice CLOSED@T7; H-ADM-COH DISCHARGED@lattice; DR2-SHARE MOOT@lattice/OPEN@arbitrary; B1 T6 on {H-LAYER,SC-SCOPE}; B5 T5.
Reproduction command:	(none -- narrative; the cited results carry the scripts)
Expected output:	n/a
Falsification gate:	Any cited result being refuted (would update this chronicle); none outstanding.
Tier before / after:	No tier action (chronicle).
No-overclaim statement:	Records cited results only. The lattice-class DR-2 and the discharge hold for the crystallographic class; arbitrary-Q DR-2 stays OPEN. B1 tier UNCHANGED T6.
Next required action:	Mainline: SC-SCOPE all-orders lift (the remaining B1 lever beyond H-LAYER).